

Cervical Cytology-Histology Correlation QA Report

Automated rule-based quality assurance analysis

Total Cases: 130

Executive Summary

A total of 130 cases were analyzed using a deterministic cytology-histology correlation QA pipeline.

Concordance Summary

Concordance_Class	Count	Percent
minor_discordant	51	39.23
major_discordant	42	32.31
concordant	37	28.46

HSIL-Focused Metrics

Metric	Value
HSIL Pap tests with histologic HSIL (CIN2/CIN3)	34.62
HSIL Pap tests with minor discrepancies	38.46
HSIL Pap tests with major false-positive discrepancies	26.92
Number of major false-negative / cytology undercall discrepancies	27.0
PV+ for HSIL Pap tests (%)	34.62

Discrepancy Bucket Summary

Bucket	Count	Percent
MinVar	4	3.08
MajUnd	27	20.77
MinUnd	31	23.85
Agree	37	28.46
MinOver	16	12.31
MajOver	15	11.54

Key Quality Signals

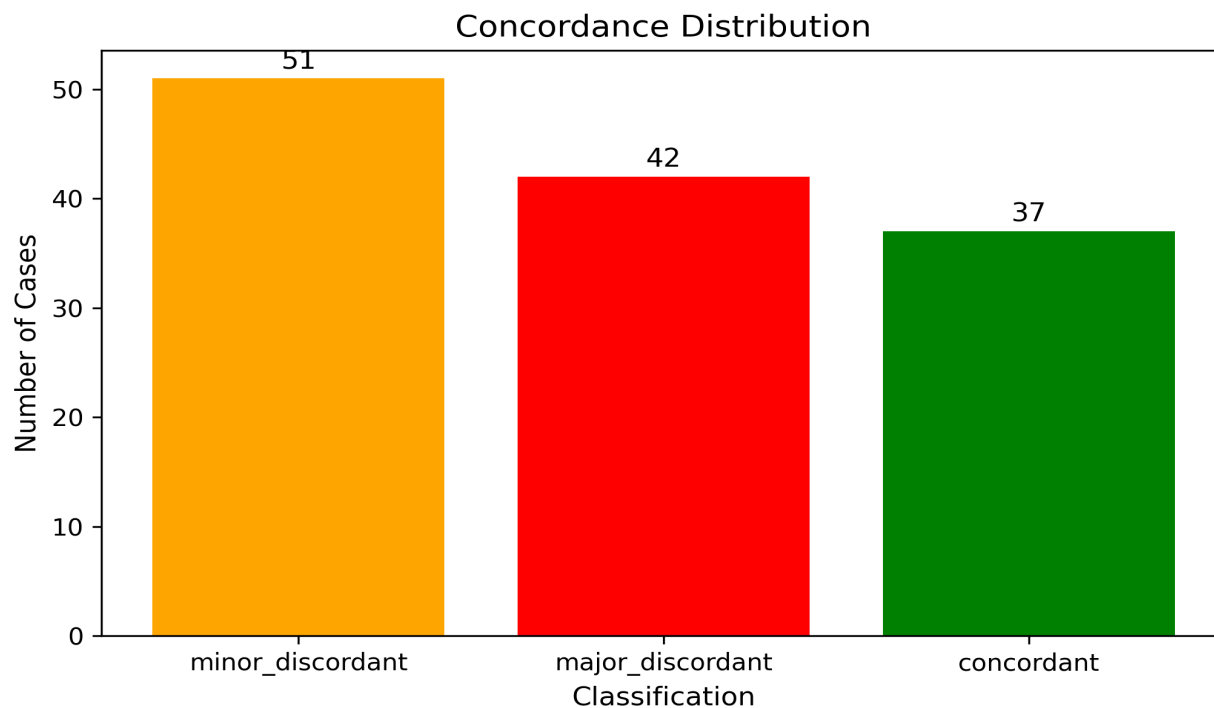
- Major discordance rate: 32.31%
- Minor discordance rate: 39.23%
- Undercall proportion: 44.62%

- Overall proportion: 23.85%

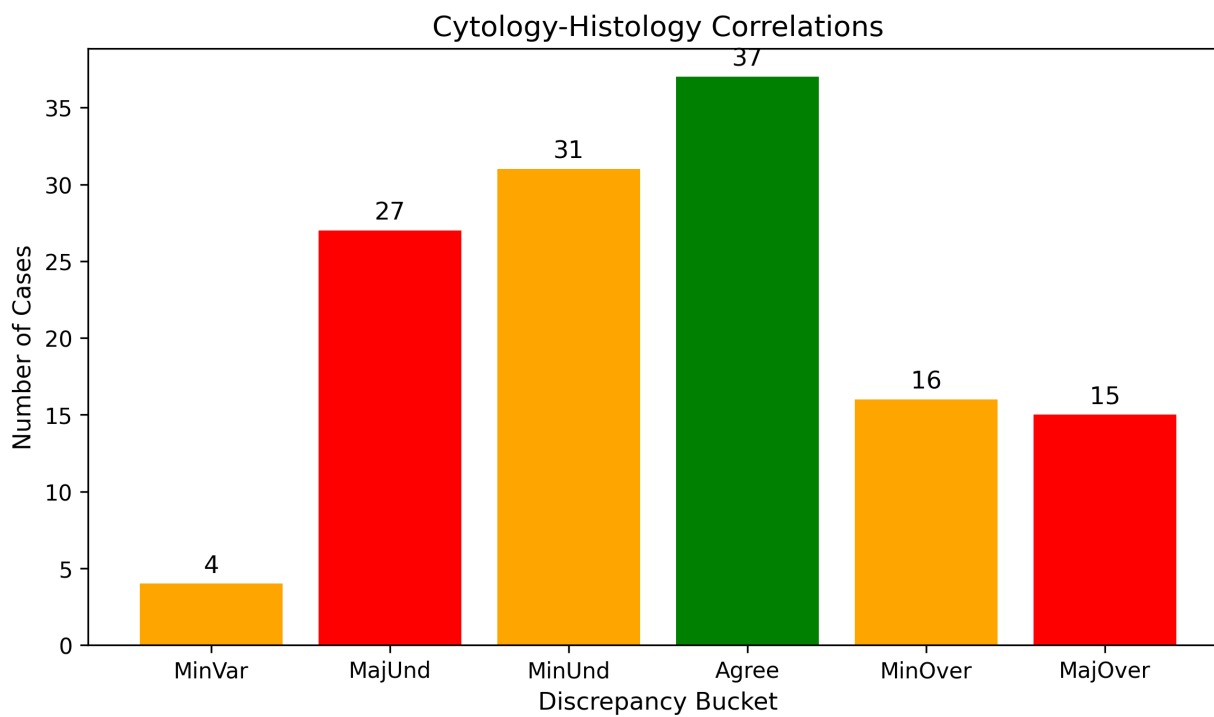
- HSIL PV+: 34.62%

Figures

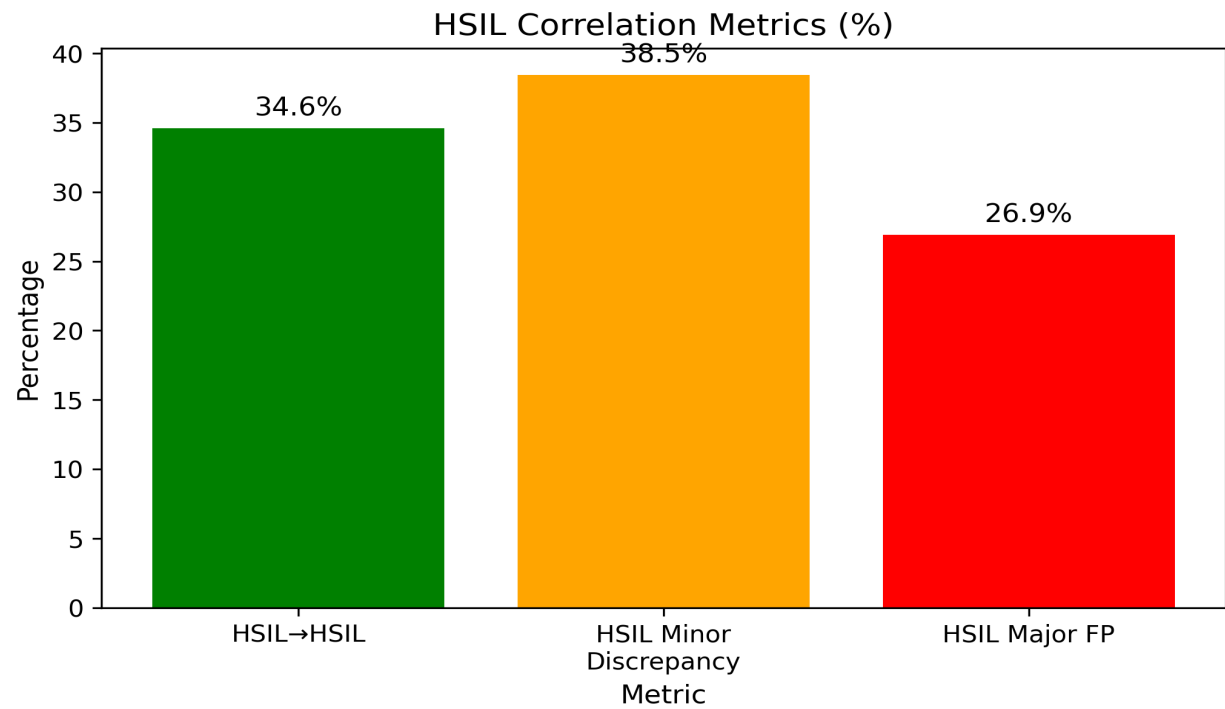
Concordance Distribution



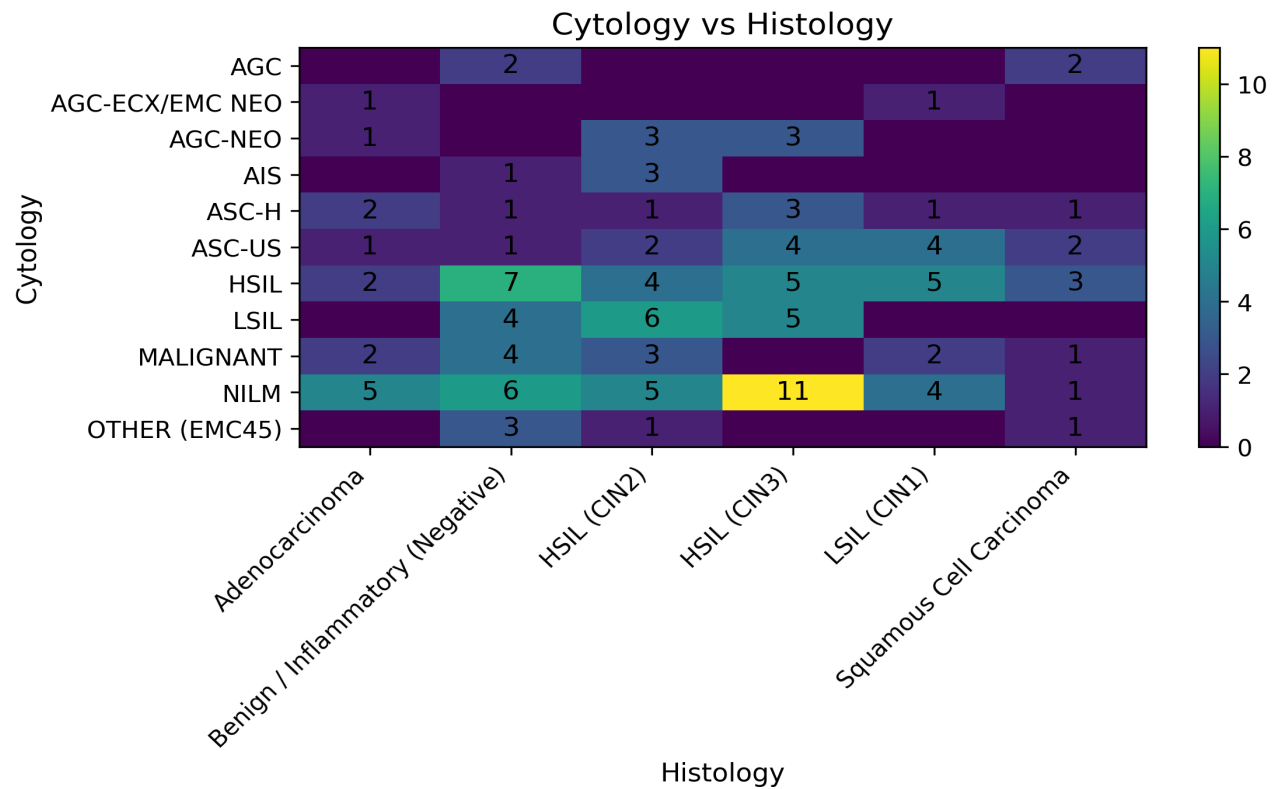
Cytology-Histology Correlation Buckets



HSIL Correlation Metrics



Confusion Matrix



Limitations

This report is based on deterministic cytology-histology correlation using rule-based classification.

Routine CHC datasets are subject to verification bias due to selective histologic follow-up.

Accordingly, these results should not be interpreted as measures of screening sensitivity or specificity.

Observed discrepancies should be interpreted within the context of laboratory QA review rather than as direct indicators of diagnostic error.